



ASSIGNMENT No: 17

TITLE: Write a Java program to demonstrate the use of Swing and Applets

Problem Statement

Write a Java program to demonstrate the use of Swing and Applets

Learning Objectives

To develop graphical user interfaces using Swing components and Applets.

Theory

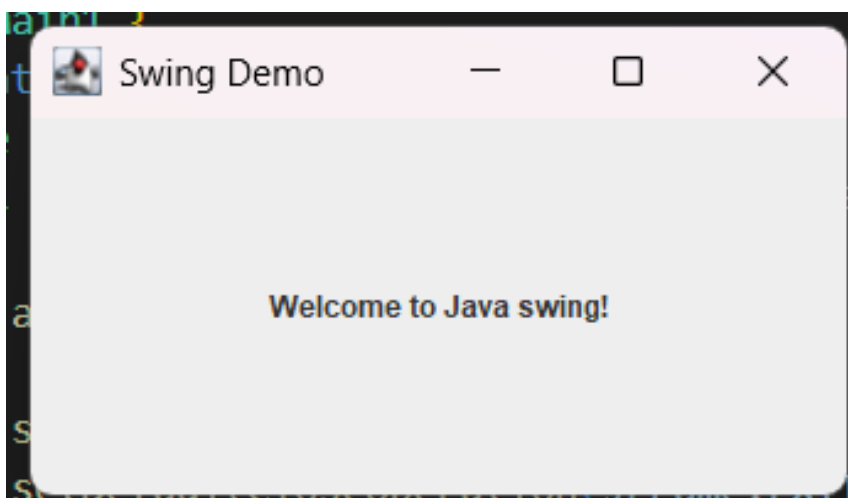
Swing is Java's GUI toolkit that provides lightweight components such as buttons, labels, text fields, and menus. Applets were small Java programs embedded in web pages for interactive applications. Although Applets are obsolete, Swing remains an important framework for desktop application development. Swing supports platform-independent graphical interfaces.

```
import javax.swing.*;

public class Main1 {
    public static void main (String[] args) {
        JFrame frame = new JFrame("Swing Demo");
        JLabel label = new JLabel("Welcome to Java swing!", JLabel.CENTER);

        frame.add(label);

        frame.setSize(350, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```



```
import java.applet.*;
import java.awt.*;

public class Main2 extends Applet {
    public void paint (Graphics g) {
        g.drawString("Welcome to Java Applet", 50, 50);
    }
}
```

Exercise

- 1 Create a simple Student Registration Form GUI using Swing.
- . Create a simple Employee Registration Form using Java Swing GUI that
- 2 takes
- . employee details such as Employee ID, Name, Department, and Salary. Display the entered information in a dialog box

```
import javax.swing.*;

public class Exp1 {
    public static void main (String[] args) {
        JFrame frame = new JFrame("Student Registration Form");

        JLabel nameLabel = new JLabel("Name:");
        JLabel rollLabel = new JLabel("Roll No:");
        JLabel courseLabel = new JLabel("Course:");

        JTextField nameField = new JTextField();
        JTextField rollField = new JTextField();
        JTextField courseField = new JTextField();

        JButton submitButton = new JButton("Submit");

        nameLabel.setBounds(50, 50, 100, 30);
        nameField.setBounds(150, 50, 200, 30);

        rollLabel.setBounds(50, 100, 100, 30);
        rollField.setBounds(150, 100, 200, 30);

        courseLabel.setBounds(50, 150, 100, 30);
        courseField.setBounds(150, 150, 200, 30);

        submitButton.setBounds(150, 210, 100, 30);

        frame.add(nameLabel);
        frame.add(nameField);
        frame.add(rollLabel);
        frame.add(rollField);
        frame.add(courseLabel);
        frame.add(courseField);
        frame.add(submitButton);

        frame.setSize(420, 320);
        frame.setLayout(null);
        frame.setVisible(true);

        submitButton.addActionListener(e -> {
            JOptionPane.showMessageDialog(
                frame,
                "Registration Successful!\nName: " + nameField.getText()
                + "\nRoll No: " + rollField.getText()
                + "\nCourse: " + courseField.getText()
            );
        });
    }
}
```

Student Registration Form

Name: Dhaarmit

Roll No: 25070122073

Course: CSE

Submit

Message

Registration Successful!

Name: Dhaarmit

Roll No: 25070122073

Course: CSE

OK

Submit

```
import javax.swing.*;

public class Exp2 {
    public static void main (String[] args) {
        JFrame frame = new JFrame("Employee Registration Form");

        JLabel idLabel = new JLabel("Employee ID:");
        JLabel nameLabel = new JLabel("Name:");
        JLabel deptLabel = new JLabel("Department:");
        JLabel salaryLabel = new JLabel("Salary:");

        JTextField idField = new JTextField();
        JTextField nameField = new JTextField();
        JTextField deptField = new JTextField();
        JTextField salaryField = new JTextField();

        JButton submitButton = new JButton("Submit");

        idLabel.setBounds(40, 40, 100, 30);
        idField.setBounds(150, 40, 200, 30);

        nameLabel.setBounds(40, 90, 100, 30);
        nameField.setBounds(150, 90, 200, 30);

        deptLabel.setBounds(40, 140, 100, 30);
        deptField.setBounds(150, 140, 200, 30);

        salaryLabel.setBounds(40, 190, 100, 30);
        salaryField.setBounds(150, 190, 200, 30);

        submitButton.setBounds(150, 240, 100, 30);

        frame.add(idLabel);
        frame.add(idField);
        frame.add(nameLabel);
        frame.add(nameField);
        frame.add(deptLabel);
        frame.add(deptField);
        frame.add(salaryLabel);
        frame.add(salaryField);
        frame.add(submitButton);
    }
}
```

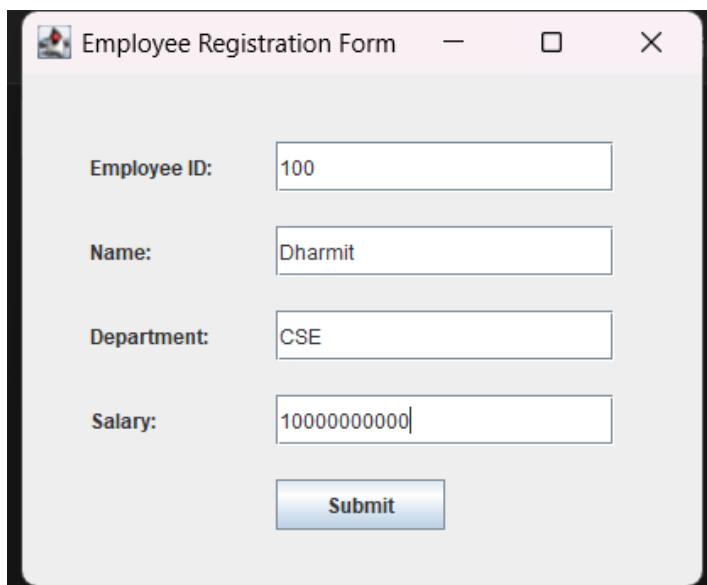
```

frame.add(idLabel);
frame.add(idField);
frame.add(nameLabel);
frame.add(nameField);
frame.add(deptLabel);
frame.add(deptField);
frame.add(salaryLabel);
frame.add(salaryField);
frame.add(submitButton);

frame.setSize(420, 350);
frame.setLayout(null);
frame.setVisible(true);

submitButton.addActionListener(e -> {
    JOptionPane.showMessageDialog(
        frame,
        "Employee Details\n"
        + "Employee ID: " + idField.getText()
        + "\nName: " + nameField.getText()
        + "\nDepartment: " + deptField.getText()
        + "\nSalary: " + salaryField.getText()
    );
});
}
}

```



Employee Registration Form

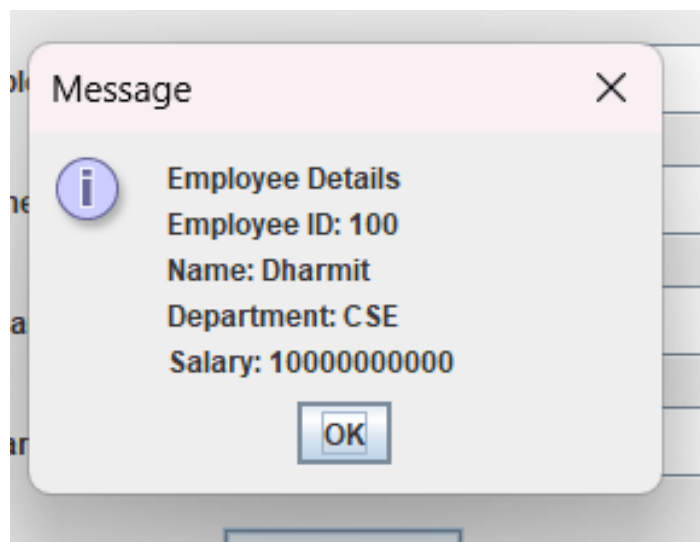
Employee ID: 100

Name: Dharmit

Department: CSE

Salary: 10000000000

Submit



Conclusion

This experiment helped in understanding the use of Java Swing for creating graphical user interfaces. A simple Employee Registration Form was developed using components such as JFrame, JLabel, JTextField, JButton, and JOptionPane. The entered employee details were successfully collected and displayed in a dialog box, demonstrating basic GUI design and event handling in Java.

